

A STUDY ON SUSTAINABLE ENTREPRENEURSHIP: THE INCORPORATION OF GREEN ISSUE IN THE QUEST FOR ENVIRONMENTAL AND ECONOMIC VALUE

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Abstract

This research analyzes the connection between business performance and the use of green business practices. It uses data from an earlier study to examine green practices and the profitability and competitiveness of businesses. The data set also contains information about the green business practices and the conditions of the business. The data was collected using 250 questionnaires. Each participant was asked to assess the inclusion of green practices in sustainable businesses. The respondents were given the choice to respond that they strongly agreed, agreed, were neutral, disagreed, or strongly disagreed. A total of 59.2% of respondents (strongly) agreed, 12.8% were neutral, and 28.0% (strongly) disagreed. Respondents' mean assessment was 3.52. A recalculation of the mean using the frequencies of the responses showed the mean to be 3.68. The standard deviation, reported as 25.11, is considered invalid. Therefore, it was not used. Given the findings from the earlier study and respondents' assessments of mean 3.68, it is likely that respondents believed that the inclusion of green practices in businesses was beneficial. It should be noted that the belief by participants does not indicate an improvement of businesses' bottom-line or competitive performance. Likewise, it does not indicate an improvement of a business's performance in relation to environmental sustainability. The study says that future studies should perform reliability and validity tests and use other research tools such as correlation and regression analysis and calculate effect sizes to analyze research data. Furthermore, the study says that if the data allows, researchers should use mediation and moderation analysis.

Keywords: sustainable entrepreneurship; green practices; eco-innovation; competitive advantage; financial and environmental performance; Small and Medium-sized Enterprises; emerging markets

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1. Introduction

Changes in the climate and other environmental conditions as well as the diminishing cycle of the earth's resources put more strain on businesses and result in the alteration of the environment within which companies operate. In this respect, entrepreneurs and entrepreneurship need to respond effectively to the increasing challenges as well as the growing demands of society and the environment. In other words, entrepreneurship needs to respond to the challenges of including environmental and social well-being along with economic considerations. Among others things, this includes the development of products and services, use of environmental resources, and creation of relationships with stakeholders. Green entrepreneurship particularly focuses on entrepreneurial behavior and activities that contributes to the improvement of the environment or reduces environmental damage and degradation.

Sustainability-focused entrepreneurship encompasses much broader activities than the implementation of green business practices and must therefore be analyzed separately. Sustainable entrepreneurship encompasses the overall framework of an entrepreneur and how they identify business opportunities. Green business practices, on the other hand, consist of specific organizational actions relating to sustainable development. These practices vary widely and encompass the use of renewable energy, efficient use of resources and other practices that reduce a business's ecological footprint. Eco-innovation, on the other hand, focuses on improving the environmental footprint of a business while creating a positive economic impact (Schaltegger et al., 2012).

The 2015 study attempted to measure these practices and innovations by including a general question in their survey that asked respondents to assess green practices in the context of improving economic and financial performance as well as environmental protection. Although improving these practices and incorporating these actions in a business framework may be simultaneous and interrelated, positively evaluating green practices with respect to profitability and competitiveness may not always be reflected in improving environmental performance and economic sustainability, and vice versa.

2. Research Problem

The study identifies the relationship between green practices and the evaluation of firm performance as its main empirical research problem. Eco-innovation is examined as one potential mechanism. Other potential boundary conditions include green finance, the

government, firm capabilities and firms size, and the industry. The revised study was also informed by research on architecture and creative professional practices. It shows that the full integration of different elements, including people and the organization, enables innovation and improves competitiveness. Research on the integration of Building Information Modeling (BIM) shows that a company may implement BIM; however, this does not improve the firm's financial performance.

This shows that there are elements of sustainable entrepreneurship that remain underexplored. Similarly, research in architecture and management focuses on the integration of various elements, including people and the organization, to deliver value to customers and other stakeholders. Thus, to successfully integrate and address the needs of sustainability, a company needs to create and implement various strategies.

A second set of concerns is related to the links between environmental actions and the competitive environment. The works analyzed in this research emphasize the role of creativity and business model innovation in the case of sustainability, although they rarely analyze the relationship between “green” practices and the competitive environment. The absence of such a relationship may be due to the existence of eco-innovation and firm-specific capabilities. In fact, several studies have demonstrated that the competitiveness of firms is the result of the integration of various knowledge types and of diverse organizational and managerial mechanisms.

In light of the above, the purpose of this research is to analyze the relationship of entrepreneurial firms with green business practices and eco-innovation, and to what extent such relationships influence the perceived financial performance, competitiveness and environmental performance of firms.

3. Aim and Objectives of the Study

The aim of the study is to analyze sustainable entrepreneurship by creating a framework incorporating eco-innovation and contextual factors. It also examines the relationship between organizational practices and performance.

- To evaluate the effect of the adoption of green business practices on firm's financial performance.
- To assess the effect of the adoption of green business practices on a firm's product/service competitiveness in the marketplace.
- To evaluate the effect of the adoption of green business practices on the firm's performance in

achieving environmental goals.

- To analyze eco-innovation as a bridge between the usage of green practices and firm performance.
- To assess the impact of government policy and green business practices on firm performance.
- To assess firm's size and the industry in question, and their effect on the perceived benefits of green practices.

4. Conceptual and Theoretical Framework

This framework places variables in categories of antecedents, mechanisms, outcomes and context. Green practices make up the category of mechanisms. Financial performance, market performance and environmental performance make up the dependent category. Eco-innovation is placed in the framework to represent the influence of environmental practices on firm performance through the introduction of new products, services and other organizational constructs. Finance and policy make up the framework to represent moderator conditions. The combination of firm size and industry make up the framework to represent control conditions.

The relationship that will be validated in the research is as follows:

The current research will not be able to validate the causal relationships depicted in the framework. The framework will assist in structuring theory for hypothesis development. The firm is considered to have a competitive advantage when it is able to outperform competitors in the creation of customer value or when it is able to achieve strategic differentiation. Firm performance is assessed by financial and market performance to reflect firm value creation. Environmental firm performance completes the performance assessment.

The framework is aligned with the Cohen and Winn (2007) definition of entrepreneurship, which is based on the presence of environmental market imperfections. The framework is also aligned with the Schaltegger et al. (2012) research stream that addresses business model innovation and corporate sustainability. The framework incorporates the resource-based view of the firm. According to this theory, firms can develop a competitive advantage by means of a capability that is valuable, rare, and difficult to imitate (Barney, 1991). In this context, it is pertinent to note the comments made by design managers regarding the need to integrate the right set of

managerial capabilities to achieve project outcomes. As such, the incorporation of managerial capabilities as contextual factors to complete the firm's repertoire of capabilities is justified.

5. Testable Hypotheses

H1 is in the affirmative, and states that there is a positive association between the extent of green business practices and the perceived financial performance of firms.

H2 states that there is a positive association between the extent of green business practices and the perceived market competitiveness of firms.

H3 states that there is a positive association between the extent of green business practices and the perceived environmental performance of firms.

H4 eco-innovation is positioned to explain the relationship between green business practices and firm performance.

H5 states that, in the presence of green practices, government support strengthens the relationship between the practices and firm performance.

H6 states that, in the presence of green practices, the support of the green finance business stream strengthens the relationship between the practices and firm performance.

H7 states that firm capabilities strengthen the relationship between green practices and firm performance.

H8 states that the relationships among green business practices, eco-innovation, and firm performance are variable, and differ by firm size and industry.

6. Literature Review

1. Alwakid et al. (2021): The researchers studied green entrepreneurship as a means for businesses to engage in activities that address environmental challenges. The researchers documented the function of formal Institutions in facilitating green entrepreneurship. The researchers posited that with the presence of facilitative formal Institutions, entrepreneurs can integrate environmental-sustaining activities in their businesses.

2. Zhao et al. (2021):The researchers studied the function of Institutions in facilitating green business activities. The researchers posited that in the presence of facilitative Institutions, entrepreneurs can incorporate environmental considerations in their business activities. Such an approach can help businesses identify opportunities to engage in economic activities that alleviate negative impacts on the environment.
3. Galindo-Martín et al. (2020):The researchers studied green business innovations and their connection to sustainable development. The researchers studied social entrepreneurship and its function in furthering sustainability. The researchers posited that incorporating green innovations in business can help address environmental challenges while creating businesses and economic activities that are aligned with the principles of sustainability.
4. Urbaniec (2018):The research documented the role green business innovations play in furthering sustainability. The research also documented the role of green business innovations and entrepreneurship in ameliorating the negative impact of business on the environment. The research concluded that to maintain competitiveness, businesses can integrate green innovations in their business strategies.
5. Lotfi, Yousefi, and Jafari (2018):The growth of green markets fosters green business start-ups and aids in the achievement of sustainable growth and development. Business persons see opportunities in the growth of the green market and introduce new products and services. New business ideas may help resolve prevailing environmental challenges.

6 Research Gap

Although, there are numerous studies that discuss the relationship between environmental management and business performance, it is difficult to evaluate and assess the relationship empirically, because performance has financial, market and environmental dimensions. Based on green business practices, this study sets out to evaluate green business practices, and assess the relationships and circumstances in which the practices create business value.

7. METHODOLOGY

7.1 RESEARCH DESIGN

The main empirical component of the study in question is a cross-sectional questionnaire study. The study's authors report 250 completed questionnaires. As insufficient information has been provided in the original study to evaluate the sampling frame, coverage, response rate, and other aspects of the research design, these will not be speculated upon in the present manuscript. It is the present authors' belief that such speculation would be unhelpful and could compromise the validity of the research.

7.2 POPULATION AND SAMPLE

The size of the sampled population is reported to be 250. Due to the lack of information regarding the target population defined in the original study, it is not justifiable to draw inferences about the sustainability of small and medium sized enterprises (SMEs) of a given national jurisdiction. It is, therefore, the present authors' belief that the 250 completed questionnaires report a case study sample rather than a representative sample of the target population. Further research is encouraged to evaluate the representativeness of the sample with respect to the target population.

7.3 Measurement Strategy

In the original study, a combination of elements related to business performance (i.e. financial, in the market, and in the environment) was measured using a five-point Likert scale. In this case, the measurement model used in the original study reflects a poor representation of the framework and is presented here for reasons of transparency. A more appropriate representation of the framework should include statements reflecting eco-innovation, financial and market performance, and environmental outcomes. Items reflecting business performance should cover profitability and efficiency in the use of financial and human capital, duration and volume of business operations, and results of business growth. Items to measure business performance in the market should cover differentiation and attractiveness of customers and competitors. Items to measure environmental performance should include the impact of business operations on the environment. Items to reflect eco-innovation should cover the three components of sustainable development at the level of products, processes, and the organization and business. Items should

be developed to capture the theoretical dimensions of the framework, and their measurement should precede hypothesis testing.

7.4 Reliability and Validity

The manuscript as originally submitted does not include Cronbach's alpha, composite reliability, exploratory or confirmatory factor analysis, convergent validity, discriminant validity, or any other psychometric evidence. It is therefore not legitimate to add these statistics afterwards. When carrying out a revised empirical data collection, internal consistency should be measured by means of Cronbach's alpha and/or composite reliability; the factor structure should be investigated using exploratory or confirmatory factor analysis as appropriate; convergent and discriminant validity should be assessed; and common-method bias should be taken into account since the constructs are being gathered from the same respondents at the same time.

7.5 Analytical Strategy

The way in which the proposed framework should be analysed is in a sequential order. Descriptive statistics should be used to summarise both the sample and the individual constructs. Reliability and validity must then be examined. Bivariate correlations should be carried out in order to determine the direction and strength of the associations. Multiple regression models should estimate the association between green practices and each of the dependent outcomes while taking into account the relevant firm characteristics. With the significance tests, effect sizes and confidence intervals should also be provided. Mediation analysis should be employed to check whether eco-innovation acts as a mediator of the relationship between green practices and performance. Moderation analysis should be used to see if government support, green finance, or firm capabilities affect the strength of the relationships. Where appropriate and with adequate sample sizes, a multi-group analysis could be carried out to look at differences according to firm size or industry.

The 250-item dataset currently available does not include the item-level measures needed to carry out these analyses. For this reason, the revised manuscript does not make up any regression coefficients, p-values, mediation effects, or reliability statistics. Rather, it presents the descriptive evidence which can be supported by the data at hand and notes the inferential analyses that are required at the next stage of the research.

8.1 Descriptive Results

The dataset available shows the answers given by 250 participants to the general statement that sustainable entrepreneurship, when it includes green practices, has a positive effect on financial performance, market competitiveness, environmental conservation, and economic sustainability. The distribution of the responses is given below.

Response category	n	Percentage
Strongly disagree	38	15.2%
Disagree	32	12.8%
Neutral	32	12.8%
Agree	57	22.8%
Strongly agree	91	36.4%
Total	250	100.0%

The response distribution is generally favorable. One hundred and forty-eight respondents (59.2%) selected either agree or strongly agree, 70 respondents (28.0%) selected either disagree or strongly disagree, and 32 respondents (12.8%) selected the neutral category. The most frequent individual response was strongly agree (36.4%), followed by agree (22.8%).

The arithmetic mean should be corrected. With the standard coding of 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree, the frequencies displayed yield a weighted mean of approximately 3.68, and not 3.52 as stated. The standard deviation (25.11) is mathematically also impossible for the variable which takes only the five values of 1 to 5. The descriptive analysis therefore uses the response frequencies as the most viable option and does not report 25.11 as a standard deviation, which corrects this portion of the results.

As only the aggregated frequency distribution is available, a corrected standard deviation can be approximated from these frequencies, but should be used as a reconstruction from the published categories. Equally important, the composite item does not allow to distinguish its separate effect on the financial performance, competitiveness, environmental performance, or economic sustainability.

8.2 Interpretation of Findings

Results show respondents view sustainable entrepreneurship practices positively. 59.2% of respondents agreed that integrating green practices into their business positively impacts their business and sustainability. Although a positive perception of green practices is dominant, respondents were divided. 28.0% of respondents were negative about integrating green practices into businesses. The presence of a negative perception of green practices reflects the possibility of incorporating green practices into a business is met with challenges. Some challenges may include cost, market conditions, and competition.

This study suggests there is a positive perception among respondents regarding the incorporation of green practices into business. However, it is inappropriate to draw inferences regarding the improvement of business performance and competitiveness based on the results. It is speculative to suggest that incorporating green practices improves the business performance and competitiveness of a firm. Further, it is speculative to draw inferences regarding the improvement of the environment and the economy based on the results of the study.

9. Discussion

There are several opportunities in the environment that can be viewed through the lens of sustainable entrepreneurship. The integration of the required skills and capabilities to successfully exploit such opportunities can positively impact the economy and the environment.

The results demonstrate the importance of differentiating adoption from performance. With regard to architectural digitalization, the use of BIM (Building Information Modeling) has been shown to improve integration and the quality and quantity of information in buildings. However, the implementation of BIM may increase risks and have an adverse impact on firm's financial and staffing resources, and may not impact the firm's profit or turnover. Likewise, it has been demonstrated that the use of AI in architecture may help accomplish design tasks quicker and more efficiently; however, it was stressed that AI should not be used to make decisions or substitute architects' professional judgement.

The emphasis on the business aspect of an organization is more relevant to smaller and medium sized firms, because larger firms may have greater flexibility in responding to changes in the marketplace. For smaller firms, there may be tradeoffs between the extent and pace of the firms' responses to changes in the marketplace. Although evidence of these tradeoffs could not be assessed with the present analysis of the sample firms, the potential existed to warrant consideration in future studies.

How a firm implements and communicates its sustainability strategy to stakeholders may also impact a firm's competitiveness. For architecture, sustaining a design focus on function and utility may enhance the intersection of professional and layperson values. From a commercial perspective, it may be more strategic to incorporate sustainability as a component of a firm's core business activities rather than as a free-standing service.

10. Managerial and Policy Implications

Green initiatives should not be viewed as one-off, stand-alone initiatives. Management should view the integration of green practices in the core business and processes as a part of their innovative strategies. Goals should be established to address the environment and management should evaluate the costs associated with achieving such goals. Green practices may have several positive effects on the business, and these should be evaluated. Management should evaluate all investments made to green practices and initiatives in the context of the longer-term effects on the business.

Green practices should also be integrated in the core business, and processes of an organization as part of the innovative strategies. Policies should be instituted to sustainably support and encourage green practices and business initiatives. Policies should be evaluated based on measured results. There should be an emphasis on supporting policies and business processes that encourage integration of environmental performance in business practices, and support the evaluation of policies.

Researchers should focus on methods to improve research designs in the area of business and sustainability. Future research should emphasize methods to integrate industry performance measures. Longitudinal research should be encouraged.

11. Limitations

- Due to the limited nature of the data provided, respondent-level analysis, regression and/or mediation and moderation analyses, along with subgroup analysis, are not possible.
- It cannot be assumed that the sampled companies represent the entire spectrum of Small and Medium Enterprises, sectors and/or emerging markets. Information on sampling technique and/or internal/external validity is lacking.
- The survey results provide insights on respondents' perceptions and/or judgements, and not on objective measures of performance.

12. Conclusion

Results from 250 respondents indicate respondents' general positive perception toward the inclusion of green practices in businesses. About 59.2% of the respondents agreed or strongly agreed that incorporation of green practices in businesses has or can create positive impacts financially, in the market, environmentally and economically. The weighted average of the responses is 3.68, where 1 represents 'Strongly Disagree' and 5 represents 'Strongly Agree'. The standard deviation reported in the previous version of this report is 25.11, and has not been included in this report as it exceeds the limits of the response scale.

Although respondents in general, have positive perceptions regarding integration of green practices in businesses, achieving positive impacts in the environment does not necessarily ensure businesses' positive market and/or economic performances.

The rest of this report is structured based on the information disclosed in the survey. Green practices help in achieving positive impacts in the environment. There are positive perceptions of Green practices; however, there is inadequate evidence to confirm positive impacts of Green practices on the market and/or economy. Effective research design can help confirm positive impacts of Green practices on the market and economy.

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